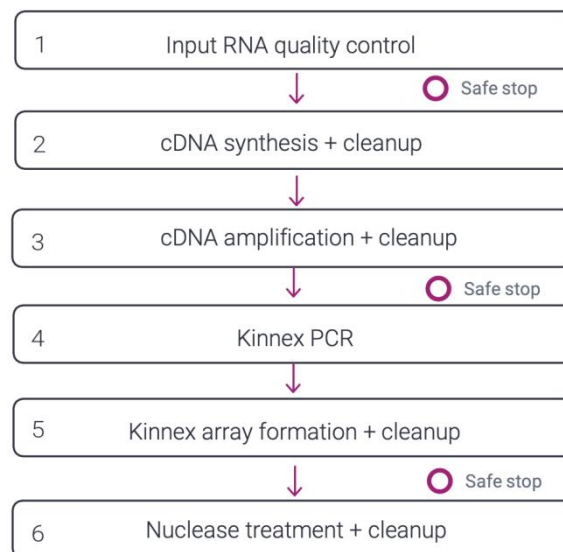


## Library construction and sequencing process

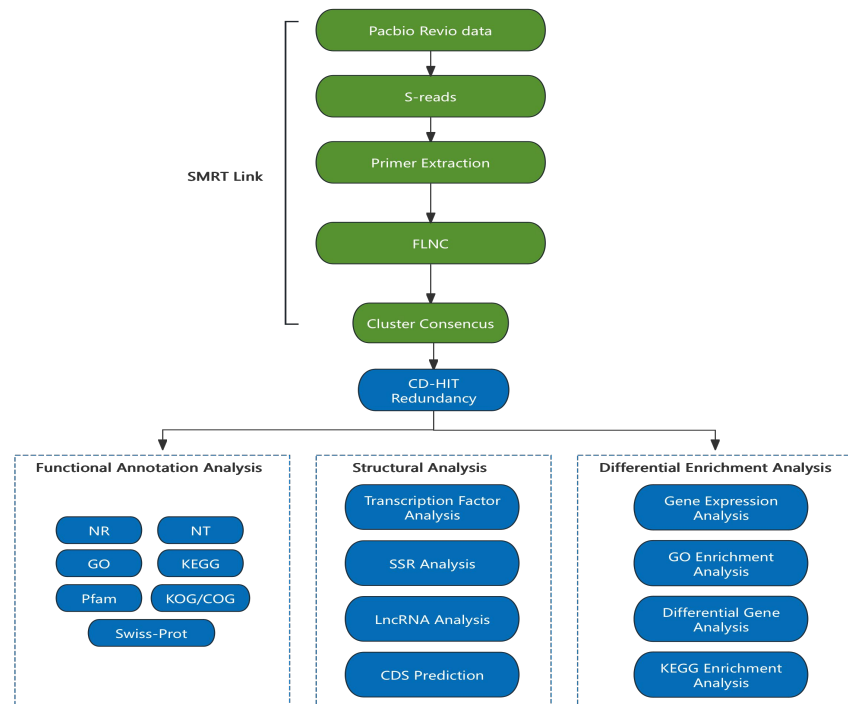
From RNA samples to final data acquisition, each link of sample testing, library construction, and sequencing will affect the quality and quantity of data, and the data quality will directly affect the results of subsequent information analysis. In order to ensure the accuracy and reliability of sequencing data from the source, Novogene strictly controls each production step of sample testing, library construction, and sequencing, fundamentally ensuring the output of high-quality data. The Kinnex full-length transcriptome library uses Iso-Seq express 2.0 kit to synthesize cDNA chains and Kinnex full-length RNA kit to build the library. The library construction process is as follows:



Official protocol: <https://www.pacb.com/wp-content/uploads/Procedure-checklist-Preparing-Kinnex-libraries-using-the-Kinnex-full-length-RNA-kit.pdf>

## Data analysis

### 1. Information analysis process



### 2. Full-length transcript analysis

After sequencing is completed, the original data will be removed from the machine for adapters and low-quality reads. The data will contain data measured with 0, 1 or 2 molecules as templates, but the information of 0 or 2 molecules will cause great interference to subsequent information analysis. This part of the data will be discarded and only the data measured with 1 molecule will be retained.

Revio platform is hifi reads after downloading, and then use software SMRTlink v 13.0 Filter and process hifi reads :

1. CCS is classified by detecting whether it contains 5' -primer, 3'-primer, and poly-A to find out FLNC (full-length non chimera) sequence and nFL (Non-Full-Length: non-full-length non-chimera sequence) sequence;
2. Cluster the FLNC sequences of the same transcript using the hierarchical  $n \cdot \log(n)$  algorithm to obtain the consensus sequence .

### 3. Full-length transcript deduplication

The corrected transcript sequences were clustered and de-redundant according to the 95% similarity between sequences , and sequence alignment and clustering were performed using CD-HIT <sup>[1]</sup> (-c 0.95 -T 6 -G 0 -aL 0.00 -aS 0.99) to remove redundant and similar sequences .

### 4. Gene function annotation

In order to obtain comprehensive gene function information, the databases used include: NR <sup>[2]</sup>, NT , Pfam, KOG/COG <sup>[3]</sup>, Swiss-prot <sup>[4]</sup>, KEGG <sup>[5]</sup>, GO <sup>[6]</sup>. The characteristics, URLs,

annotation methods and parameters of these seven databases are as follows:

| database   | Introduction                                                                               | More Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Software and parameters                           |
|------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Nr         | NCBI non-redundant protein sequences, NCBI official protein sequence database              | It includes protein coding sequences of GenBank genes, PDB (Protein Data Bank) protein database, SwissProt protein sequences, and protein sequences from databases such as PIR (Protein Information resource) and PRF (Protein Research Foundation). Its characteristics are that the content is relatively comprehensive, and the annotation results will contain species information, which can be used for species classification.                                                                                                                                                                                                                                                                                                                                                     | diamond v0.8.36; e-value = 1e-5, --more-sensitive |
| Nt         | NCBI nucleotide sequences, NCBI official nucleic acid sequence database                    | Includes nucleotide sequences from GenBank, EMBL and DDBJ (but not EST, STS, GSS, WGS, TSA, PAT, HTG sequences).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ncbi-blast-2.7.1+; e-value = 1e-5                 |
| Pfam       | Protein family, the most comprehensive classification system for protein domain annotation | Proteins are composed of domains, and the protein sequence of each specific domain is conservative to a certain extent. PFAM divides the domains of proteins into different protein families, and establishes the HMM statistical model of the amino acid sequence of each family by comparing the protein sequences. PFAM families are divided into two categories according to the reliability of the annotation results: the Pfam-A family with high reliability of manual annotation and the Pfam-B family automatically generated by the program. Through the HMMER3 program, you can search for the HMM model of the established protein domain, so as to annotate the protein family of Gene. For details, see <a href="http://pfam.sanger.ac.uk/">http://pfam.sanger.ac.uk/</a> . | HMMER 3.1 package, hmmscan; --acc                 |
| KOG/COG    | COG: Clusters of Orthologous Groups of proteins; KOG: euKaryotic Ortholog Groups           | Both KOG and COG are based on gene orthologous relationships in NCBI, where COG is for prokaryotes and KOG is for eukaryotes. COG/KOG divides homologous genes from different species into different ortholog clusters based on evolutionary relationships. Currently, COG has 4873 classifications and KOG has 4852 classifications. Genes from the same ortholog have the same function, so the functional annotations can be directly inherited to other members of the same COG/KOG cluster. For details, see <a href="http://www.ncbi.nlm.nih.gov/COG/">http://www.ncbi.nlm.nih.gov/COG/</a> .                                                                                                                                                                                       | diamond v0.8.36; e-value = 1e-5, --more-sensitive |
| Swiss-Prot | A manually annotated and reviewed protein sequence database                                | A collection of protein sequences that have been collated and studied by experienced biologists. For details, see <a href="http://www.ebi.ac.uk/uniprot/">http://www.ebi.ac.uk/uniprot/</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | diamond v0.8.36; e-value = 1e-5, --more-sensitive |

|      |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                          |
|------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| KEGG | Kyoto Encyclopedia of Genes and Genomes                                                           | <p>A database that systematically analyzes the metabolic pathways of gene products and compounds in cells and the functions of these gene products. It integrates data on genomes, chemical molecules, and biochemical systems, including metabolic pathways (KEGG PATHWAY), drugs (KEGG DRUG), diseases (KEGG DISEASE), functional models (KEGG MODULE), gene sequences (KEGG GENES), and genomes (KEGG GENOME). The KO (KEGG ORTHOLOG) system links various KEGG annotation systems together. KEGG has established a complete KO annotation system that can complete the functional annotation of the genome or transcriptome of newly sequenced species. For details, see <a href="http://www.genome.jp/kegg/">http://www.genome.jp/kegg/</a>.</p> | diamond v0.8.36; e-value = 1e-5, --more-sensitive                        |
| GO   | Gene Ontology, an internationally standardized classification system for describing gene function | <p>GO is divided into three categories of ontologies: biological process, molecular function and cellular component, which are used to describe the biological process, molecular function and cellular environment of the products encoded by genes. The basic unit of GO is term, each term has a unique identifier (consisting of "GO:" plus 7 numbers, such as GO:0072669); the terms of each category of ontology form a directed acyclic topological structure through the connections between them (is_a, part_of, regulate). For details, please see <a href="http://www.geneontology.org/">http://www.geneontology.org/</a>.</p>                                                                                                             | Self-written script based on protein annotation results of Pfam database |

## 5. Gene structure analysis

### 6.1 CDS Prediction

We used the ANGEL software to perform CDS prediction analysis<sup>[7]</sup>. The software includes two modes: error-free and error-tolerant (error-tolerant mode is used by default). When implementing prediction in error-tolerant mode, ANGLE mainly uses a machine learning algorithm that makes the most of the limited information from the input sequence to predict the coding sequence: it not only uses the codon usage frequency, but also uses protein structure information that is difficult to apply to random model algorithms. In this way, limited information can be optimized when predicting the coding potential of short sequences, so the accuracy of the prediction results does not depend on the length of the input sequence.

### 6.2 Transcription factor analysis

Animal transcription factors were identified using the animal transcription factor database<sup>[8]</sup>-animalTFDB 2.0. For species included in the database, if the gene is from Ensembl geneid, transcription factors were directly screened. For genes that are not from Ensembl geneid, BLASTX screening was performed with the known transcription factor protein sequences of the

species in the database. For species not included in the database, hmmsearch was used to identify them based on the pfam file of the transcription factor family. Plant transcription factors were identified using iTAK 1.7a (-f 3F).

### 6.3 SSR Analysis

In our analysis, MISA software (version 1.0, default parameters) was used; the corresponding minimum number of repeats for each unit size were: 1-10, 2-6, 3-5, 4-5, 5-5, 6-5 (e.g., 1-10, when a single nucleotide is used as the repeating unit, the repeat number must be at least 10 to be detected; 2-6, when a double nucleotide is used as the repeating unit, the minimum number of repeats is 6) to perform SSR detection on Gene, [for details, see] <http://pgrc.ipk-gatersleben.de/misa/misa.html>.

### 6.4 LncRNA Analysis

Due to the limitation of library construction principle, we could only obtain lncRNAs with poly A tails. We used CNCI<sup>[9]</sup>, PLEK<sup>[10]</sup>, CPC<sup>[11]</sup> software and Pfam<sup>[12]</sup> database to predict the coding potential of PacBio sequencing data, and then conducted subsequent analysis on the obtained lncRNAs.

## 6. Gene expression level analysis

CD-HIT software was used to remove redundancy from the corrected consensus sequence as a "gene". The readcount value and TPM value of each "gene" were directly obtained through the clustering file and the correspondence file between flnc and consensus sequences.

## 7. Gap Analysis

The input data for differentially expressed gene analysis is the readcount data obtained from the gene expression level analysis. The analysis is mainly divided into three parts:

1. First, normalize the readcount;
2. Then the hypothesis test probability (pvalue) is calculated based on the model;
3. Finally, multiple hypothesis testing correction was performed to obtain the FDR value (false discovery rate).

Different software will be used to analyze differentially expressed genes in different situations. The analysis methods are as follows:

| type                      | software                    | Standardized methods | pvalue calculation model       | FDR calculation method | Differential gene screening criteria                      |
|---------------------------|-----------------------------|----------------------|--------------------------------|------------------------|-----------------------------------------------------------|
| Biological replication    | DESeq2 (Anders et al, 2014) | DESeq                | Negative binomial distribution | BH                     | $ \log_2(\text{FoldChange})  > 0$ & $\text{padj} < 0.05$  |
| No biological replication | DEGseq (Wang et al, 2010)   | TMM                  | Poisson distribution           | BH                     | $ \log_2(\text{FoldChange})  > 1$ & $\text{padj} < 0.005$ |

## 8. GO enrichment analysis of differentially expressed genes

We used Goseq software for GO enrichment analysis, which is based on the Wallenius non-central hyper-geometric distribution relative to the ordinary hyper-geometric distribution. The characteristics of this distribution are that the probability of extracting an individual from a certain category is different, and this difference in probability is obtained by estimating the preference for gene length, so that the enrichment of GO terms by differential genes can be calculated more accurately. Probability.

## 9. KEGG enrichment analysis of differentially expressed genes

KOBAS v3.0 (Corrected P-Value < 0.05) was used for KEGG enrichment analysis. The integrated metabolic pathway query provided by KEGG is excellent, including the metabolism of carbohydrates, nucleosides, amino acids, etc. and the biodegradation of organic matter. It not only provides all possible metabolic pathways, but also comprehensively annotates the enzymes that catalyze each step of the reaction, including amino acid sequences, links to PDB libraries, etc. It is a powerful tool for in vivo metabolic analysis and metabolic network research.

## 10. How to decompress the result file

| Compression format                         | User Type            | method                                                 |
|--------------------------------------------|----------------------|--------------------------------------------------------|
| Compressed file in the form of file.tar.gz | Unix/Linux/Mac users | Use tar -zxvf file.tar.gz command                      |
|                                            | Windows Users        | Use decompression software such as WinRAR, 7-Zip, etc. |
| Compressed file in file.gz format          | Unix/Linux/Mac users | Use gzip -d file.gz command                            |
|                                            | Windows Users        | Use decompression software such as WinRAR, 7-Zip, etc. |
| Compressed file in file.zip format         | Unix/Linux/Mac users | Use unzip file.zip command                             |
|                                            | Windows Users        | Use decompression software such as WinRAR, 7-Zip, etc. |

## 11. Result file format description

| File Type     | File Description                                                                                                                    | Open                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| file.fa/fasta | Sequence file, fasta format, usually gene sequence or genome sequence. Because the file is generally large, it is difficult to open | Unix/Linux/Mac users use the less or more command                  |
|               |                                                                                                                                     | Windows users use advanced text editors such as Editplus/Notepad++ |
| file.fq/fastq | Sequence file, fastq format, usually reads sequence; because the file is generally large, it is difficult to open                   | Unix/Linux/Mac users use the less or more command                  |
|               |                                                                                                                                     | Windows users use advanced text editors such as Editplus/Notepad++ |

|              |                                                                                                                                                                                                |                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| file.txt/xls | Result data table file; the file is separated by tabs                                                                                                                                          | Unix/Linux/Mac users use the less or more command                                                    |
|              |                                                                                                                                                                                                | Windows users can use advanced text editors such as Editplus/Notepad++, or open with Microsoft Excel |
| file.pdf/svg | Result image file; vector image, can be enlarged or reduced without distortion, convenient for users to view and edit, can be edited with Adobe Illustrator, used for article publishing, etc. | Windows/Mac users can use Adobe Reader/Foxit Reader/web browser to open                              |
|              |                                                                                                                                                                                                | Unix/Linux users use the evince command to open                                                      |
| file.png     | Resulting image file; bitmap, lossless compression                                                                                                                                             | Unix/Linux/Mac users use the display command to open                                                 |
|              |                                                                                                                                                                                                | Windows users can use image browsers to open, such as Photoshop, etc.                                |

## 12. Analysis software list and versions

| Analyze content                  | Software and version    | parameter                                                                                                   | Remark                                                                | Software Links                                                                                                |
|----------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Raw data processing and analysis | SMRTlink V 13.0         | default                                                                                                     | Full-length transcript sequence identification and redundancy removal | <a href="https://www.pacb.com/support/software-downloads">https://www.pacb.com/support/software-downloads</a> |
| CDS                              | ANGEL V2.4              | --min_angel_aa_length 50                                                                                    | CDS Forecast                                                          | <a href="https://github.com/PacificBiosciences/ANGEL">https://github.com/PacificBiosciences/ANGEL</a>         |
| TF                               | Plants:iTAK:1.7a        | -f 3F                                                                                                       | Transcription factor identification                                   | <a href="https://github.com/kentnf/iTAK/">https://github.com/kentnf/iTAK/</a>                                 |
|                                  | Animal: AnimalTFDB: 2.0 | Default Parameters                                                                                          | Transcription factor identification                                   | <a href="http://bioinfo.life.hust.edu.cn/AnimalTFDB/">http://bioinfo.life.hust.edu.cn/AnimalTFDB/</a>         |
| SSR                              | MISA 1                  | definition(unit_size, min_repeats):1-10 2-6 3-5 4-5 5-5 6-5; interruptions (max_difference for_2_SSRs): 100 | SSR identification                                                    | <a href="http://pgrc.ipk-gatersleben.de/misa/misa.html">http://pgrc.ipk-gatersleben.de/misa/misa.html</a>     |



|                                  |                |                                                 |                             |                                                                                                                                                     |
|----------------------------------|----------------|-------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| lncRNA analysis                  | CPC2 v0.1      | Default Parameters                              | Coding potential prediction | <a href="http://cpc2.cbi.pku.edu.cn/download.php">http://cpc2.cbi.pku.edu.cn/download.php</a>                                                       |
|                                  | CNCI V2        | Default Parameters                              | Coding potential prediction | <a href="https://github.com/www-bioinfo-org/CNCI">https://github.com/www-bioinfo-org/CNCI</a>                                                       |
|                                  | PLEK v1.2      | Default Parameters                              | Coding potential prediction | <a href="https://sourceforge.net/projects/plek/">https://sourceforge.net/projects/plek/</a>                                                         |
|                                  | PfamScan V1.6  | Default Parameters                              | Protein domain detection    | <a href="https://www.ebi.ac.uk/seqdb/confluence/display/THD/PfamScan">https://www.ebi.ac.uk/seqdb/confluence/display/THD/PfamScan</a>               |
| Transcript redundancy removal    | cd-hit v4.6.8  | -c 0.95; -T 6; -G 0; -aL 0.00; -aS 0.99; -AS 30 | Transcript deduplication    | <a href="https://github.com/weizhongli/cdhit">https://github.com/weizhongli/cdhit</a>                                                               |
| Differential expression analysis | DESeq v1.10.1  | qvalue < 0.05                                   | Biological replication      | <a href="http://www.bioconductor.org/packages/release/bioc/html/DESeq.html">http://www.bioconductor.org/packages/release/bioc/html/DESeq.html</a>   |
|                                  | DEGseq v1.12.0 | log2(FoldChange)  > 1 & qvalue < 0.005          | No biological replication   | <a href="http://www.bioconductor.org/packages/release/bioc/html/DEGseq.html">http://www.bioconductor.org/packages/release/bioc/html/DEGseq.html</a> |
| GO enrichment                    | GOSec V1.10.0  | Corrected P-Value<0.05                          | GO enrichment analysis      | <a href="http://www.bioconductor.org/packages/release/bioc/html/goseq.html">http://www.bioconductor.org/packages/release/bioc/html/goseq.html</a>   |
| KEGG enrichment                  | KOBAS v3.0     | Corrected P-Value<0.05                          | KEGG enrichment analysis    | <a href="http://kobas.cbi.pku.edu.cn/download.php">http://kobas.cbi.pku.edu.cn/download.php</a>                                                     |

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